MILITARY SPECIFICATION

GUIDE, SHARPENING, DENTAL INSTRUMENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a sharpening guide suitable for use in sharpening dental hand instruments.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitations for bid or requests for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited).
QQ-N-290	Nickel Plating (Electrodeposited).
MMM-A-250	Adhesive, Water-Resistant, (For Sealing Fiberboard Boxes).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper (for Carton Sealing).

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

FSC 6520

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DRAWINGS

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

DPSC Drawing 24124 Guide, Sharpening, Dental Instrument.

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bid or requests for proposal shall apply.

U.S. DEPARTMENT OF COMMERCE - NATIONAL BUREAU OF STANDARDS

CS 271 Grading of Abrasive Grain for Grinding Wheels.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality throughout and shall be suitable in all respects for the intended purpose.

3.1.1 Sheath. The sheath shall be fabricated from brass, chromium over nickel plated. The sheath shall be plated with nickel conforming to class 1, type VI (KC) of QQ-N-290, followed by a plating of chromium conforming to class 1, type I of QQ-C-320.

3.1.2 Hone.

* 3.1.2.1 Abrasive head. The abrasive head shall be fabricated from grit number 240 high quality aluminum oxide abrasive bonded with high quality vitrified ball clay ceramic binder. The abrasive grain size shall be in accordance with the limits for sizing of U. S. Department of Commerce Commercial Standard CS271 for Grading of Abrasive Grain for Grinding Wheels.

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3.1.2.2 Cementing agent. The cementing agent for mounting the abrasive head on the mandrel shall be an epoxy or phenolic resin cement suitable for the purpose.

3.1.2.3 Mandrel. The mandrel shall be fabricated from carbon or alloy steel, chromium or nickel plated, or from AISI 416 corrosion-resistant steel.

3.1.3 Other components. The adjustable table, sleeve, collar, thumbscrew nut, support pin, and swivel arm shall be fabricated from carbon or alloy steel plated with copper and nickel conforming to class 1, type III (KS) of QQ-N-290, followed by a final plating of chromium conforming to class 1, type I of QQ-C-320.

3.2 Style, design, and dimensions. The instrument sharpening guide shall be suitable for use in the dental operatory for sharpening dental hand instruments in a precise and controlled manner, when attached to a conventional, Doriot type, round-nosed dental handpiece. The guide shall consist of a sheath, a swivel arm, an offset arm (sleeve, collar, thumbscrew nut, and support pin), an adjustable table rest, and two mandrel-mounted hones. The guide shall conform to Defense Personnel Support Center (DPSC) Drawing 24124 for style, design, and dimensions.

3.2.1 Sheath. The sheath shall fit over the nose of a Doriot type, round-nosed straight handpiece in the same manner as a short, round sleeve contra-angle. The sheath shall be provided with a recess to receive a mandrel-mounted hone, and the distal bearing surface of the sheath shall support the mandrel.

3.2.2 Mandrel-mounted hones. Each guide shall be provided with a large and a small mandrel-mounted hone. The mandrel shall extend through the mesial end of the sheath a sufficient distance to permit firm clamping in the handpiece bur chuck, whether the mandrel is fully or partially inserted into the sheath.

3.2.2.1 Large hone. The large hone shall be suitable for use in sharpening the larger dental hand instruments, such as dental chisels, enamel hatchets, periodontal hoes (including the side bevels on these instruments), gingival margin trimmers, angle formers, periodontal knives, sickle and wing scalers, lancets, gold knives, bi-beveled hatchets, spoon excavators, and small scissors.

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3.2.2.2 Small hone. The small hone shall be suitable for use in sharpening smaller dental hand instruments, such as Jaquette scalers, Gracey curettes, and small spoon excavators.

3.2.3 Swivel arm. The swivel arm shall properly fit the distal end of the sheath and the proximal end of the offset arm, and shall serve as the means of connection between the sheath and the offset arm.

3.2.4 Offset arm. The offset arm shall be composed of a support pin, over which a sleeve and collar fit, and a knurled thumbscrew nut. The arm shall receive the adjustable table rest, and shall be capable of locking the table rest firmly into the required position by turning the thumbscrew nut clockwise.

3.2.5 Adjustable table rest. The table rest shall be T-shaped, and in profile shall resemble the shape of an inverted, truncated cone. The table rest shall be adjustable up and down relative to the stone, and shall be adjustable a full 360 degrees around the cutting face or distal surface of the hone, to provide any desired bevel on a dental hand-cutting instrument.

3.3 Runout. The abrasive head shall be trued to size and shape. The total indicated runout shall not exceed 0.006 inch when measured as specified in 4.4.1.

3.4 Operational test. The guide shall comply with the operational test specified in 4.4.2.

3.5 Finish. The exposed surfaces of the guides shall have a mirror finish.

3.6 Instructions. Adequate and accurate instructions for operation and maintenance shall be included with each guide.

3.7 Identification marking. Each guide shall be permanently and legibly stamped with the name or registered trademark of the manufacturer.

3.8 Workmanship. Workmanship shall be first class throughout. Guides shall be free from defects which detract from their appearance, or may impair their serviceability, proper functioning, and intended use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

* 4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the material, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

* 4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in tables I and III (see 4.2.1 and 4.3) and table II (see 4.2.2).

* 4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table I. Unit of product for examination purposes shall be one sharpening guide.

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Table I. Inspection and Acceptable Quality Levels

Type defect	Inspection level	AOL (Percent defective)
Major	II	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for end item testing. Sampling for tests on end item shall be conducted in accordance with MIL-STD-105 and as indicated in table II. Acceptance number shall be zero for all sample sizes. The unit of product shall be one sharpening guide.

Table II. Tests

Characteristic	Requirement	Test procedure	Inspection level
Plating	3.1.3	QQ-C-320 QQ-N-290	S-1
Runout	3.3	4.4.1	S-2
Operational test	3.4	4.4.2	S-2

4.2.3 Sampling for raw material testing. A sufficient quantity from each lot of aluminum oxide abrasive shall be selected for determination of abrasive grain size.

* 4.3 Examination. The sharpening guides shall be examined for defects, including but not limited to, those listed in table III.

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Table III. Classification of Defects

Category	Defect
Major:	
101	Style and design of sharpening guide not as specified.
102	Any component of sharpening guide missing or broken.
103	Finish not mirror.
104	Instructions not present with sharpening guide.
105	Workmanship defects such as chips or nicks present.
Minor A:	
201	Identification marking not present; not permanent and legible.
202	Packaging and packing not as specified.
203	Marking not as specified.
Minor B:	
251	Dimensions not as specified.

4.4 Tests.

4.4.1 Runout. The total indicated runout of the hones shall be measured to the nearest 0.001 inch. The hone shall be held in a precision chuck and rotated slowly. A dial indicator or other suitable equipment shall be employed to measure the runout of the mounted head. The difference between the minimum and maximum readings for one full rotation of the wheel shall be taken as the total indicated runout.

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4.4.2 Operational test. The guide shall be attached to a round nose Doriot type straight handpiece and used as directed in the accompanying instructions for sharpening a smaller and a larger cutting type periodontal hand instrument. The guide shall be tested for ability to properly guide and sharpen the instruments.

5. PREPARATION FOR DELIVERY

* 5.1 Packaging. Packaging shall be level A, or C, as specified (see 6.1).

* 5.1.1 Unit of issue. One guide, sharpening, dental instrument, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	144 units

5.1.2.1 Variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a rigid plastic box with a clear plastic hinge cover. Box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Closure shall be adequate to prevent accidental opening.

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5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2-inch wide tape conforming to PPP-T-76. Closing of bottom flaps with either adhesive (conforming to MMM-A-250) or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers, of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

* 5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A.

* 5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and shall be constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.

* 5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and shall be constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers, of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

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5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

* 6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

* 6.2 Specification coverage. This specification covers the requirements for FSN 6520-935-5880, GUIDE, SHARPENING, DENTAL INSTRUMENT.

* 6.3 Supersession data. This specification supersedes Purchase Description No. 1 dated 4 May 1968.

* 6.4 Specification changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.5 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-G-36651 FSN:

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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